

Microeconomic Theory: TA Session

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About this TA Section

Some logistics:

- ▶ Fridays, 9:00-9:50am, Gilman 132
- ▶ What we'll do: review class material; go over assignments; answer any questions; do practice problems...
- ▶ Slides will be made available after each session on Github:
https://github.com/pindawang/IntermediateMicro_Fall26
(I have sent a Canvas announcement containing this link)

My office hours:

- ▶ In-person: Wednesdays, 2:30pm-3:30pm, Wyman Park Building W601A
- ▶ Zoom: by appointment
- ▶ My email: pwang66@jhu.edu

Homework 1, Q1

Professor Barbara likes his coffee sweet. He states that he strictly prefers cup-of-coffee A to cup-of-coffee B if A has more than $1/4$ teaspoon more sugar than B. If two cups of coffee have a sugar difference of $1/4$ teaspoon or less he cannot taste the difference between them and is indifferent between them.

1. Are his stated weak preferences complete?
2. Are his states weak preferences transitive?

Answer:

1. Yes. Any two cups of coffee are comparable.
2. No. Counter-example: three cups of coffee: A has 0.5 teaspoon of sugar, B has 0.7 teaspoon, and C has 0.9 teaspoon. We have $A \succsim B$ and $B \succsim C$, but $C \not\succsim A$. Preferences are not transitive.

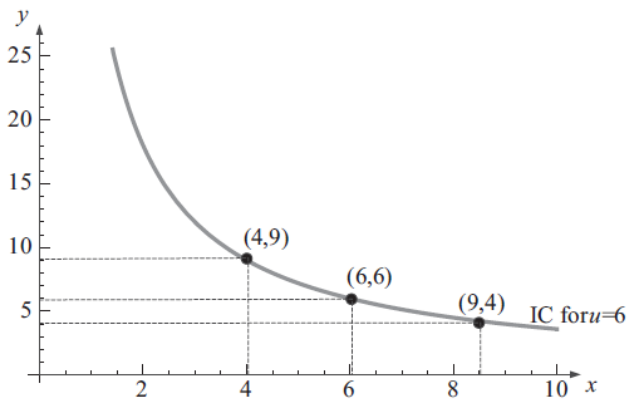
Utility functions

Suppose preferences are represented by $u(x, y) = \min\{x, y\}$.

1. Are preferences strictly monotonic?
2. Are preferences monotonic?

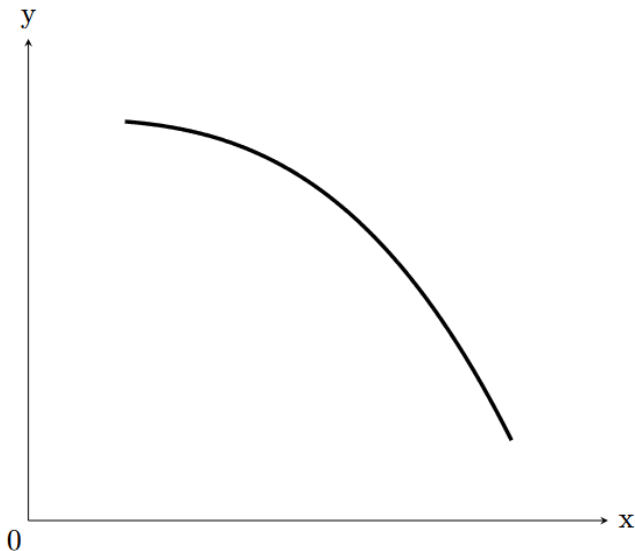
Answer: (1) No. (2) Yes.

Indifference curves



Indifference curves

Typical indifference curves are convex. What does an indifference curve that is not convex look like?



Indifference curves

Why can't a typical indifference curve be thick? Which property of preferences does it violate?

Answer: If an indifference curve is thick, then there exist bundles (x_1, y_1) and (x_2, y_2) such that $x_1 > x_2$ and $y_1 > y_2$ but $(x_1, y_1) \sim (x_2, y_2)$. This violates monotonicity.

Homework 1, Q2

Consider the following utility functions over mangos and nectarines, $(m, n) \geq (0, 0)$:

1. $u_i(m, n) = 2m + 3n$
2. $u_{ii}(m, n) = 16m^2 + 24mn + 9n^2$
3. $u_{iii}(m, n) = 8m + 12n - 24$

Two of these utility functions represent the same preferences. (a) Which two? Explain. (b) For the other utility function, find a comparison (i.e. (m_1, n_1) vs (m_2, n_2)) on which the two preferences disagree.

Answer:

(a) $u_{ii}(m, n) = (4m + 3n)^2$ is not a strictly increasing transformation of $u_i(m, n) = 2m + 3n$, whereas $u_{iii}(m, n) = 4(2m + 3n) - 24$ is a strictly increasing transformation of $u_i(m, n)$. So, the first and third utility functions represent the same preferences.

(b) The two preferences disagree on $(m_1, n_1) = (1, 0)$ and $(m_2, n_2) = (0, 1)$.

Transformations of utility functions

Strictly increasing transformations preserve utility rankings

- ▶ Suppose $u(x, y)$ is a utility function, and $f(u)$ is a strictly increasing function, then the utility function $v(x, y) = f(u(x, y))$ represents the same preferences as $u(x, y)$

Examples of strictly increasing $f(u)$:

- ▶ Linear transformations: $f(u) = 3u - 5$
- ▶ Exponential transformations: $f(u) = 2u^2$ (assuming utilities are non-negative)
- ▶ Log transformations: $f(u) = \log u$

Transformations of utility functions

Consider the following two utility functions:

1. $u = 4x + 2y$
2. $v = 4x^2 + y^2 + 4xy + 2x + y$

Do they represent the same preferences?

Answer: Yes, because $v = \frac{1}{4}(4x + 2y)^2 + \frac{1}{2}(4x + 2y)$; it's a strictly increasing transformation of u .

Transformations of utility functions

Consider the following two utility functions:

1. $u = x + y$
2. $v = (x + y)^2 + 3x$

Do they represent the same preferences?

Answer: No, because it's not a transformation of u . The rankings of (4,0) and (0,5) are different with these two utility functions. $v = (x + y)^2 + 3(x + y)$ would have been a strictly increasing transformation of u , but $v = (x + y)^2 + 3x$ is not.

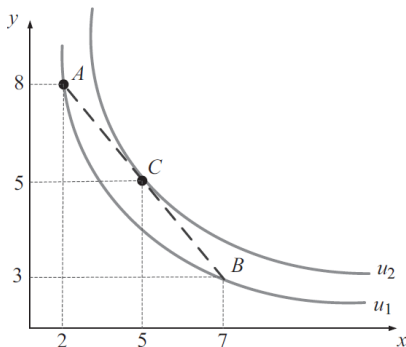
Marginal rate of substitution

Marginal rate of substitution: $MRS_{xy} = \frac{MU_x}{MU_y}$

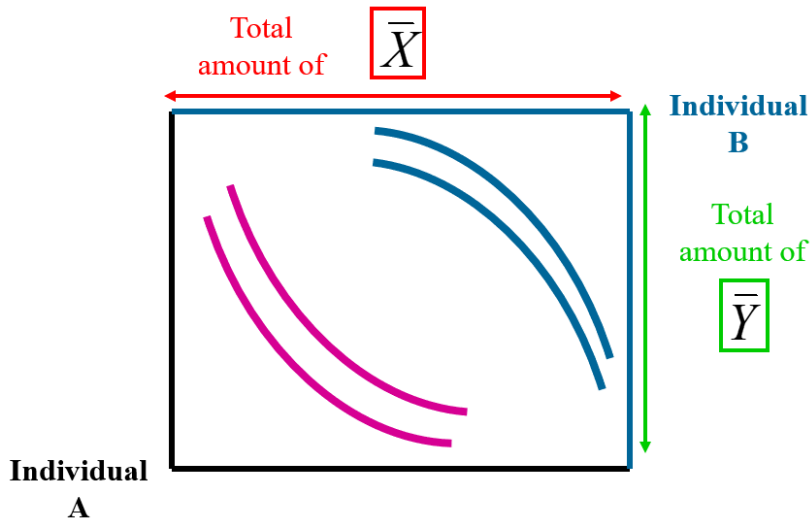
Graphically, MRS is the absolute value of the slope of the indifference curve.

- In the graph below, MRS at point C is 1

Convexity $\Leftrightarrow$ diminishing MRS



Edgeworth box



Pareto efficiency

A **Pareto improvement** is a change that makes at least one person better off without making anyone else worse off

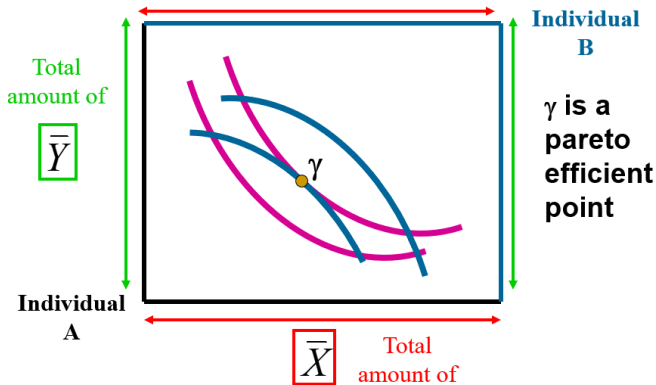
An allocation is **Pareto efficient** if it is impossible to make someone better off without making anyone else worse off

Pareto efficiency

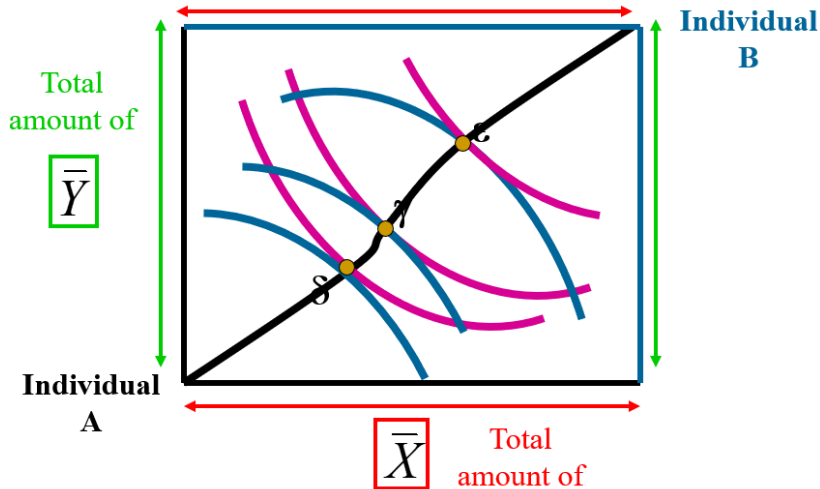
If an allocation is not on the edges, indifference curves of the two individuals must be **tangent** at the point of a Pareto efficient allocation:

$$MRS_A = MRS_B$$

- ▶ Otherwise we can always move “inward” to find a point which lies above both consumers’ original indifference curves



Contract curve



Contract curve is the set of all Pareto efficient allocations; it is the black curve in the graph above

Pareto efficiency - exercise

There are two consumers A and B, and two goods x and y . The total available amount of both goods is 4. The utility of consumer A is $U_A = x_A^{3/4} y_A^{1/4}$; the utility of consumer B is $U_B = x_B^{1/4} y_B^{3/4}$.

1. Suppose A consumes $(x_A, y_A) = (4, 0)$ and B consumes $(x_B, y_B) = (0, 4)$. Is this Pareto efficient?

Answer: No. With this allocation, both consumers have a utility of 0. Any other interior allocation (e.g. A consumes (2,2) and B consumes (2,2)) is a Pareto improvement upon this.

2. Suppose A consumes (4,4) and B consumes (0,0). Is this Pareto efficient?

Answer: Yes. Any other allocation will make consumer A worse off.

Pareto efficiency - exercise

There are two consumers A and B, and two goods x and y . The total available amount of both goods is 4. The utility of consumer A is

$U_A = x_A^{3/4} y_A^{1/4}$; the utility of consumer B is $U_B = x_B^{1/4} y_B^{3/4}$.

3. Find a Pareto efficient allocation that is in the interior (i.e. not at the edges of the Edgeworth box).

Answer: $MRS_A = \frac{\partial U_A / \partial x_A}{\partial U_A / \partial y_A} = \frac{3y_A}{x_A}$; $MRS_B = \frac{\partial U_B / \partial x_B}{\partial U_B / \partial y_B} = \frac{y_B}{3x_B}$.

With a Pareto efficient allocation, it must be that $MRS_A = MRS_B$
 $\Rightarrow \frac{3y_A}{x_A} = \frac{y_B}{3x_B}$. A convenient solution is $(x_A, y_A) = (3, 1)$, and
 $(x_B, y_B) = (1, 3)$. Another example of a solution would be
 $(x_A, y_A) = (2, \frac{2}{5})$, and $(x_B, y_B) = (2, \frac{18}{5})$.

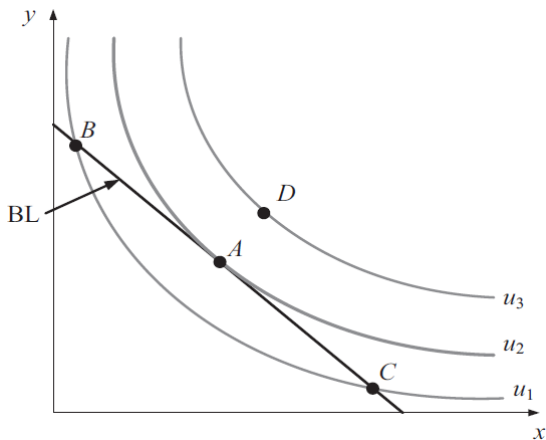
4. Derive the contract curve in terms of x_A and y_A .

Answer: From (3), we have $\frac{3y_A}{x_A} = \frac{y_B}{3x_B}$. We also know that
 $x_B = 4 - x_A$, $y_B = 4 - y_A$. Plugging these into the MRS equation,
we have $\frac{3y_A}{x_A} = \frac{4-y_A}{3(4-x_A)} \Rightarrow y_A = \frac{x_A}{9-2x_A}$, $0 \leq x_A \leq 4$, $0 \leq y_A \leq 4$.

Utility maximization

Given a budget constraint, the consumer's utility is maximized when the indifference curve just touches the budget constraint

► In math, $MRS_{xy} = \frac{MU_x}{MU_y} = \frac{p_x}{p_y}$



Utility maximization - exercise

1. John has the following utility function for apples (x) and bananas (y): $U = 3x^{0.5}y^{0.5}$. He has 20 dollars. One apple costs \$4, and one banana costs \$2. Solve for John's optimal bundle.

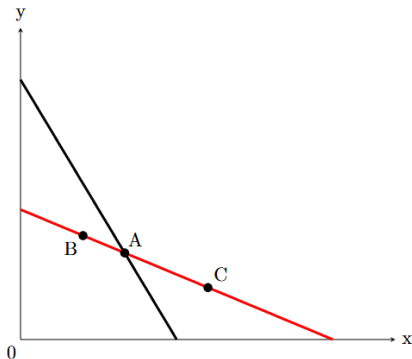
Answer: Using $MRS_{xy} = \frac{MU_x}{MU_y} = \frac{p_x}{p_y}$:

$\frac{1.5x^{-0.5}y^{0.5}}{1.5x^{0.5}y^{-0.5}} = \frac{4}{2} \Rightarrow \frac{y}{x} = 2$. Plugging into the budget constraint:

$$4x + 2y = 4x + 2 \cdot 2x = 20 \Rightarrow x = \frac{5}{2}, y = 5$$

Revealed preference

Suppose that the consumer's utility is maximized at point A when his budget constraint is the black line. Now his budget constraint shifts to the red line. Which point(s) among A, B, and C can possibly be his new utility maximization bundle?



Answer: A and C. From the consumer's choice under the black budget line, we know he prefers A over B. Under the new red budget line, A and B are still feasible, so he cannot possibly have B as the utility maximization bundle.